

Shafiya Kausar

AI Engineer · Data Scientist · Machine Learning Engineer

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LinkedIn | GitHub | Portfolio | Medium

Profile

AI Engineer and Data Scientist specialising in multimodal AI, clinical machine learning, and LLM reliability. Published author with peer-reviewed conference presentations and two book chapters in production, alongside experience building production systems across medical imaging, EHR, and physiological signal data. Seeking full time opportunities applying multimodal foundation models to real problems.

Education

MSc AI & Data Science — Aivancity, Paris 2024 – 2026

BSc Physics, Chemistry & Mathematics — Bengaluru North University, India 2020 – 2023

Publications

Agentic Artificial Intelligence for Preventive Healthcare and Early Diagnosis: Architectures, Algorithms, and Human-Centric Clinical Decision Systems In Press

- Book chapter in *Agentic AI for Sustainable Solutions in the Delivery of Healthcare* (Elsevier). *Accepted*

Agentic AI for Predictive Analytics in Industry 5.0: A Technical Review of Algorithms and Use Cases Across Healthcare, Manufacturing, Education, and Sustainable Engineering In Press

- Book chapter in *Empowering Autonomy: The Shift from Generative to Agentic AI in Collective Intelligence Systems* (CRC Press, Taylor & Francis). *Accepted*

When LLM Signals Hurt: A Coverage-Density Analysis of LLM-Augmented Reinforcement Learning for Stock Trading 2026

- Sole-authored preprint on SSRN and SocArXiv (OSF). Multi-seed evaluation of an LLM-augmented PPO-CVaR agent with signal-coverage sweep and per-regime decomposition. [Code](#).

Conference Presentations

IANLP 2026 — Int’l Conf. on AI & NLP, Faculté des Sciences Ben M’Sick, Hassan II University of Casablanca (online) 29–30/06/2026

- **Oral presentation** at the inaugural edition of IANLP: “PhenoPrompt: A Prompt-Based Clinical Phenotyping Framework Using Entity Extraction, Unsupervised Clustering, and Retrieval-Augmented Generation.” [Code](#) · [live demo](#).

PyConES 2026 — Universitat de Barcelona, Spain (in person) 06/11/2026

- Accepted **90-minute workshop** (with Anuradha Kar): “Detecting LLM Hallucinations with Uncertainty Fingerprinting: A Python Implementation.” [Code](#) · [live demo](#).

Selected Research Projects

OncoTwin — Cancer Patient Digital Twin [🔗](#)

- Probabilistic patient digital twin fusing medical imaging with structured clinical data to forecast disease progression and simulate treatment strategies, with a 3D interface.

Medical AI Dashboard [🔗](#)

- Streamlit app for medical imaging: DICOM exploration, chest X-ray classification (ResNet50 vs. ViT, ~0.95 AUC) with Grad-CAM, and 3D organ segmentation via MONAI U-Net.

PhenoPrompt — Prompt-Based Clinical Phenotyping [🔗](#)

- Query a phenotype space built from clinical notes in natural language: medkit NER → HDBSCAN clustering → entity-augmented RAG retrieval. Live demo. *Presented at IANLP 2026*.

MIMIC-II Disease Phenotyping [🔗](#)

- Unsupervised phenotyping of ICU patients (PCA, UMAP, KMeans) with an interactive Streamlit dashboard and CI pipeline. Foundation for the PhenoPrompt framework above.

WorldMedQA × BiomedCLIP [🔗](#)

- Zero-shot multimodal medical question-answering with Microsoft’s BiomedCLIP vision-language model on the 568-question WorldMedQA-V benchmark.

FedHealth — Federated Learning [🔗](#)

- Privacy-preserving federated learning (PyTorch FedAvg) for heart-disease risk prediction across simulated hospitals, with a tested library and Streamlit UI.

Metacognitive Uncertainty Layer [🔗](#)

- Hallucination detection via self-consistency, semantic entropy, logit calibration, and NLI contradiction

checking, evaluated across three models and four datasets (incl. MedQA). Fingerprint-aware adaptive policy raises accuracy from 76.5% to 93.9% on TriviaQA at 81.5% coverage. *Basis of the PyConES 2026 workshop.*

AADMN — Graph-Based Detection Re-scoring

- YOLOv8n + GNN/GAT ablation study for detection re-scoring in crowded scenes on CrowdHuman. *Pre-publication.*

Climate Displacement Research Agent

- Production research agent: hybrid RAG (BM25 + dense + RRF), cross-encoder reranking, custom MCP server, layered guardrails, and a critic agent.

Skills

Technical: Python, R, SQL, Flask

ML/AI: Deep Learning, NLP, Multimodal AI, Vision-Language Models, RAG, Time-Series, Federated Learning, Document Intelligence

Libraries: PyTorch, TensorFlow, MONAI, Scikit-learn, Pandas, NumPy, Hugging Face, Streamlit

Tools: Power BI, Git, MLflow, Docker, Google Cloud, Azure

Languages: English (Fluent), French (A1)

Professional Experience

AI Developer — SPIE ICS (Client: INSEAD, Fontainebleau)  12/2025 – Present

- Built an agentic RAG knowledge base unifying administration, rules, code, payments, and HR documentation.
- Developed AI agents automating annual resource planning (replacing a manual Google Sheets workflow) and scheduling the on-site AV/IT support team via Outlook and ServiceNow.

Research Assistant — Aivancity School for Technology, Business & Society, Paris  01/2025 – 01/2026

- Developed pipelines for extracting text, images, and tables from PDFs; optimized multi-zone document intelligence systems for structured data extraction.

Data Scientist — IntHealth, Paris 05/2025 – 01/2026

- Built mood-to-genre NLP pipeline, increasing user engagement by 35%.
- Delivered production-ready MVP integrating sentiment analysis with recommendations.

Data Scientist — AURA, Paris  01/2025 – 04/2025

- Built CNN/RNN architectures for ECG data, improving seizure detection accuracy by 17%.
- Automated preprocessing pipelines, increasing data throughput by 25%.

Research Assistant (AI/NLP) — NeuralTeks, Paris  09/2024 – 12/2024

- Integrated multimodal retrieval models achieving 0.82 mAP.
- Improved embeddings via prompt optimization and semantic alignment.

Machine Learning Engineer Intern — Codveda  07/2025 – 09/2025

- Improved model performance by 12% through benchmarking and evaluation optimization; streamlined evaluation workflows and metric consistency.

Product Manager Intern — Progton, Bangalore  06/2025 – 09/2025

- Defined and prioritized the SaaS analytics roadmap, reducing delivery time by 20%; analyzed 1000+ user feedback entries to improve usability and product alignment.

Assistant Manager Operations / Data Intern — AR Cars, Bangalore 10/2022 – 11/2022

- Built Excel and Power BI dashboards for costs, logistics, and procurement tracking; increased sales by 37% through improved stakeholder insight.

Volunteer Experience

UN Volunteer — UI/UX Designer, Action Lab for Development, Canada  01/2026 – 05/2026

- UX testing of the MoralityCode ethical AI app: identified 11 critical bugs and usability issues across the user journey, and designed Before/After mockups with implementation specs covering input validation, navigation, and flow.

Food Volunteer — Linkee  & Secours Populaire Français, Paris  02/2024 – 12/2025

- Collect and distribute meals and surplus supermarket food to people in need and refugees across Paris, through Linkee and the national solidarity network.

Certifications

AI Product Engineering (UC Berkeley SCET) · Digital Transformation in Supply Chains (PMI) · Creating API Documentation (LinkedIn Learning)